Gen-AI + Agentic AI with JavaScript (Gemini Edition)

A 10-week syllabus for MERN developers. One topic per day, about 1.5 to 2 hours. Every day follows the same format: Concept (English + Hinglish), Workflow, Code, Quiz.

Stack: Node.js, Express, MongoDB, React, Google Gemini API (free tier)

Ground Rules

- Use the `@google/genai` package. The older `@google/generative-ai` package is legacy, so avoid tutorials that use it.
- Free tier is best used with Flash / Flash-Lite models. Pro models are mostly unavailable on free tier. Check the exact limits for your account in Google AI Studio.
- Free tier data may be used by Google to improve models, so never send private or sensitive data.
- Keep the model name in one `config.js` file because model names change often.
- Type the code yourself, do not copy-paste. Push to GitHub every day.

Suggested Repo Structure

```
/phase-1-foundations/day-03-tokens
/phase-3-rag/day-20-first-rag
/projects/rag-chat-app
/projects/support-agent
```

Phase 0: Setup (Day 1-2)

Day	Topic
1	Gen-AI landscape: AI vs ML vs LLM, what Gen-AI is, job roles (AI Engineer vs ML Engineer)
2	Repo setup, `.env`, API key safety, Google AI Studio tour, first API call with `@google/genai`

Phase 1: LLM Foundations (Day 3-9)

Day	Topic
3	How an LLM works: tokens, context window, next-token prediction
4	Parameters: temperature, topP, topK, maxOutputTokens
5	System instructions, roles, multi-turn chat
6	Streaming responses (SSE) with Express + React
7	Structured output: JSON mode + schema (Zod)
8	Error handling: 429 errors, retries, exponential backoff, timeouts
9	**Mini Project 1:** CLI chatbot + streaming chat UI

Phase 2: Prompt Engineering (Day 10-14)

Day	Topic
10	Prompt anatomy: zero-shot and few-shot
11	Chain-of-thought, thinking models, reasoning
12	Prompt templates, versioning, prompt injection basics
13	Multimodal input: image, PDF, audio
14	**Mini Project 2:** Resume analyzer (PDF upload to structured JSON)

Phase 3: Embeddings, Vector DB and RAG (Day 15-26)

Day	Topic
15	Embeddings: turning text into numbers, cosine similarity
16	Semantic search with the Gemini embedding model (in-memory)
17	Chunking strategies: size, overlap, recursive splitting
18	Vector DB basics, MongoDB Atlas Vector Search
19	Ingestion pipeline: PDF/Docs, chunks, embeddings, DB
20	Retrieval + generation: your first RAG
21	Citations, showing sources, reducing hallucination
22	Advanced RAG: re-ranking, hybrid search, query rewriting, metadata filters
23	Chat memory: short-term vs long-term, summarization
24	LangChain.js basics (`@langchain/google-genai`): loaders, splitters, chains
25	Vercel AI SDK (`@ai-sdk/google`): `streamText` , `generateObject` , `useChat`
26	**Project 3:** "Chat with your Docs" full MERN RAG app

Phase 4: Agentic AI Core (Day 27-40)

Day	Topic
27	What is an agent? LLM vs Workflow vs Agent

Day	Topic
28	Function / tool calling in Gemini (most important topic)
29	Write the agent loop yourself, no framework: think, act, observe
30	Real tools: weather API, DB query, web search, calculator
31	Tool design best practices, validation, error handling
32	Agent patterns: ReAct, Plan-and-Execute, Reflection
33	Workflow patterns: prompt chaining, routing, parallelization
34	LangGraph.js: nodes, edges, state
35	LangGraph.js: conditional edges, loops, checkpointing
36	Human-in-the-loop: approval before actions
37	Multi-agent systems: supervisor, handoffs
38	Agent memory + RAG as a tool (Agentic RAG)
39	MCP (Model Context Protocol): concept + build your own MCP server
40	**Project 4:** AI Agent (Support Agent or Research Agent) with tools + memory

Phase 5: Advanced Gemini Features (Day 41-45)

Day	Topic
41	Context caching, token counting, cost awareness
42	Grounding with Google Search, URL context

Day	Topic
43	Code execution tool
44	Live API: real-time voice/video (overview + demo)
45	Image understanding and generation use cases

Phase 6: Production Engineering (Day 46-56)

Day	Topic
46	Evaluation: test sets, LLM-as-judge, RAG eval metrics
47	Guardrails: input/output validation, PII, prompt injection defense
48	Observability: logging, tracing (Langfuse / LangSmith free tier)
49	Rate limiting, queues (BullMQ), caching (Redis)
50	Security: API key protection, auth, abuse prevention
51	Cost and latency optimization, model routing
52	Testing AI apps (Vitest/Jest with a mocked LLM)
53	Deployment: Docker, Vercel/Render, env management
54	Provider-agnostic design (swap Gemini for another model easily)
55-56	Revision + architecture review

Phase 7: Portfolio and Job Prep (Day 57-70)

- **Capstone:** Full-stack MERN AI SaaS with auth, RAG, an agent, streaming UI, and a live deployment

- README with architecture diagram, demo video, and live link
- Resume bullets, LinkedIn and GitHub polish
- Interview prep: How does RAG work? How do you reduce hallucination? Agent vs workflow? System design for AI apps
- Mock interview questions (also added to the daily quizzes)

Progress Tracker

- [ ] Phase 0: Setup
- [ ] Phase 1: LLM Foundations
- [ ] Phase 2: Prompt Engineering
- [ ] Phase 3: Embeddings, Vector DB and RAG
- [ ] Phase 4: Agentic AI Core
- [ ] Phase 5: Advanced Gemini Features
- [ ] Phase 6: Production Engineering
- [ ] Phase 7: Portfolio and Job Prep

Projects Checklist

- [ ] Project 1: CLI chatbot + streaming chat UI
- [ ] Project 2: Resume analyzer
- [ ] Project 3: Chat with your Docs (MERN RAG)
- [ ] Project 4: AI Agent with tools + memory
- [ ] Capstone: MERN AI SaaS, deployed